

## Surveying in RPD

### Learning objectives

- Define surveying in RPD manufacturing
- Distinguish parts of the dental surveyor.
- Explain the principles and rules of surveying.
- Describe the objectives of surveying

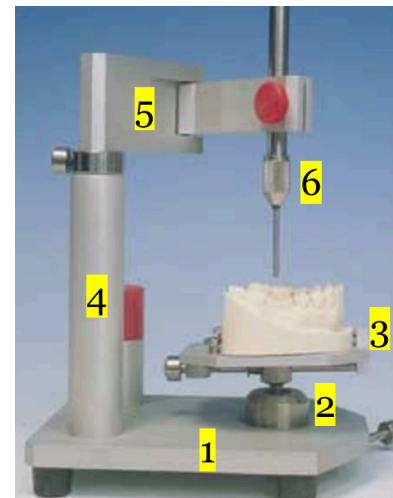
**Surveying:** The procedure of locating or delineating the contour and position of the abutment teeth and associated structures before designing a removable partial denture

**Surveyor:** An instrument used in the construction of a removable partial denture to locate and delineate the contours and relative positions of abutment teeth and associated structures.

The surveyor is a parallelometer, an instrument used to determine the relative parallelism of surfaces of teeth or other areas on a cast.

### Parts of a Surveyor

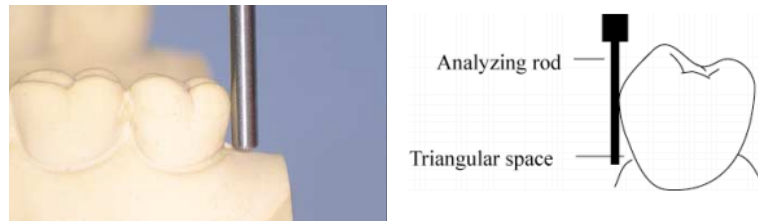
1. Platform: on which the base is moved
2. Horizontal surveyor base: on which the table swivels
3. Table or cast holder: to which the cast is attached
4. Vertical arm: It arises vertically from the surveying platform. It supports the superstructure
5. Horizontal arm: It extends horizontally from the top of the vertical arm. It is designed to support the surveying arm.
6. Surveying arm: for holding surveying tools



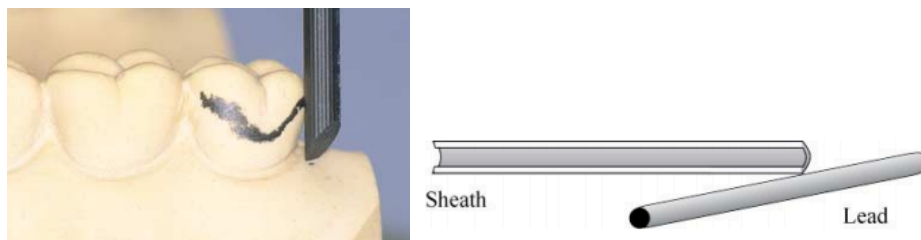
## Surveying tools

Surveying tools are attached to the surveying arm and are of four types

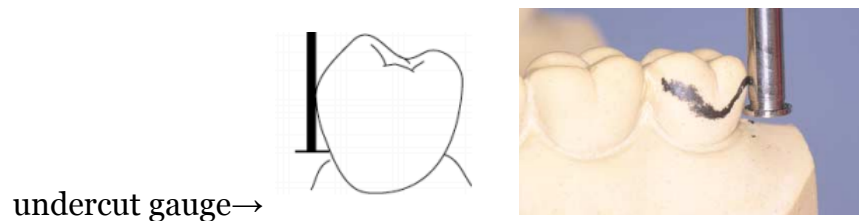
1. **Analyzing rod:** It is a rigid metal rod used for diagnostic purposes in the selection of the path of placement and to determine the undercut areas before scribing the height of the contour with the carbon marker.



2. **Carbon marker:** It is used to mark the survey line on the cast. It is protected from breakage by a protective sheath.



3. **Undercut gauge:** These gauges measure the extent of the undercut on abutment teeth used for clasp retention.

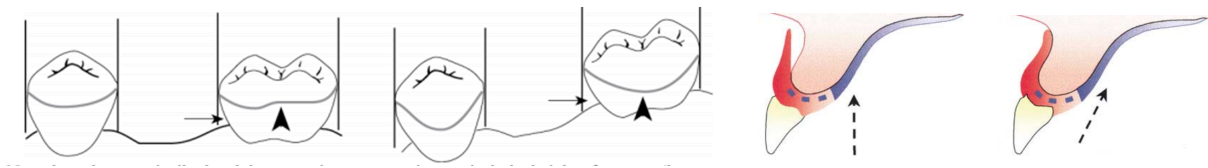


4. **Wax trimmer:** It is used to trim excess wax that may be inserted into undercut areas to be obliterated to obtain the proper form of the wax pattern.



## Objectives of Surveying

1. To determine the most desirable path of insertion and removal.
2. To locate and measure areas of the teeth that may be used for retention.
3. To determine the most suitable path of placement that will permit the best esthetic advantage.
4. To determine whether the tooth and bony areas of interference will need to be eliminated surgically or by selecting a different path of placement.
5. To delineate the height of contour on abutment teeth and to locate areas of undesirable tooth undercuts, to be avoided, eliminated, or blocked out.
6. To permit an accurate charting of the mouth preparation to be made (rest seats and guiding planes).



## Principle of surveying:

Surveying a tooth consisted of accurately locating the height of its maximum contour in relation to the plane in which the cast is positioned. The majority of natural tooth crowns are rounded (convex) in shape and have a bulge area which could occur anywhere between the occlusal surface and the gingival margin.

When the vertical analysing arm comes into contact with the convex surface, it contacts only at one point, which is the point of maximum convexity. When this surface is rotated, and is still in contact with the vertical analysing arm, an imaginary line will be traced at the greatest circumference. When we substitute this vertical analysing rod with a carbon marker, an actual line is produced at the level of the maximum tooth bulge; this line is called the height of contour (survey line). The area of a tooth occlusal to the survey line is a non-undercut area, while the area gingival to the survey line is an undercut area.

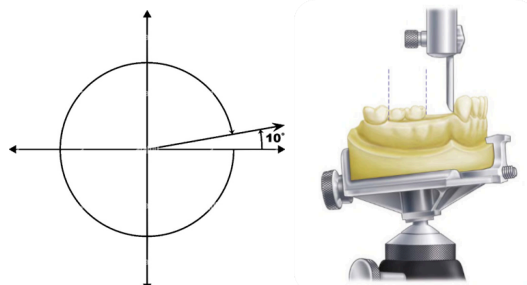


When a tooth is tilted or rotated in relation to the analyzing rod, another survey line will be traced; as a result, the extent of the non-undercut area and the undercut area are consequently changed. The survey line can vary according to the angle formed by the contact of the vertical analysing rod with the tooth surface. The undercut area can be altered by anterior and posterior tilting of the dental cast. So the effect of tilting a cast on the surveyor will be:

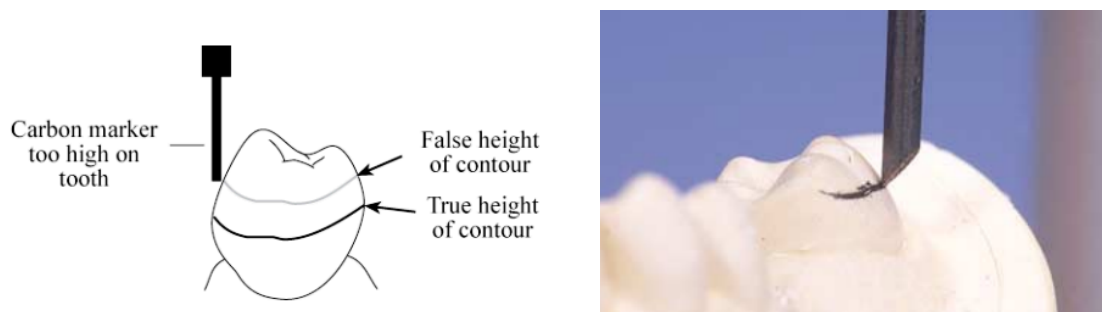
1. Redistribution of undercuts to the desired areas.
2. Allow a more favourable path of insertion.
3. Allow the use of a design to minimize food entrapment and plaque accumulation (minimize the undesirable undercut).

### Rules of surveying:

- Place the cast on the surveyor table and orient the plane of occlusion relatively horizontally. The final tilt of the cast for the ideal path of insertion is seldom more than  $10^\circ$  from this position.



- A false survey line will be produced if the tip of the marker is incorrectly positioned

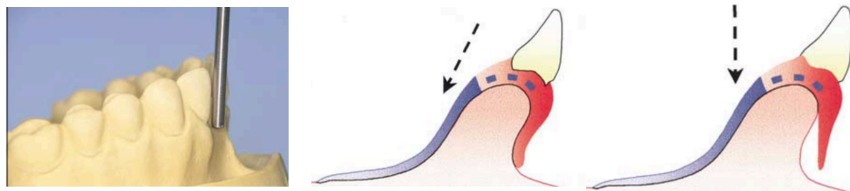
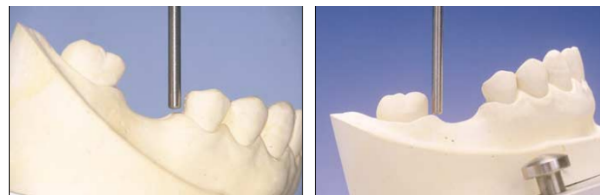


- The retentive tip of the clasp must engage the undercut area, which is present when the cast is surveyed in a certain position.



In the graph, since only the terminal end of the retentive arm flexes (circled area), the clasp on the left will lock on the cast. The rigid portion of the clasp will not be released from the undercut. The clasp on the right will release since only the flexible tip is in the undercut.

- **Anteroposterior tilt:** The anterior tilt will increase the mesial undercut, while the posterior tilt will increase the distal undercut. Such as in free-end extension partial dentures. Tilting the cast anteriorly will decrease or eliminate the distal undercut, where the path of insertion will be changed, thus getting rid of the undesirable undercut located distally. Therefore, the tilting of the cast is to minimise or equalise the undesirable undercut.



- **Lateral tilt:** dealing with a retentive undercut situated buccally or lingually on posterior teeth.

